

Customer Shopping Behavior Analysis

Key Findings and Business Recommendations

1. Purpose

The goal of this analysis was to understand customer spending behavior, product performance, discount usage, subscription patterns, and sales trends. Python was used to clean and prepare the dataset, SQL was used to answer business questions, and Power BI was used to create a dashboard for visual analysis.

2. Key Findings

Finding 1: Male customers generated most of the revenue

The SQL results showed that male customers generated **\$157,890**, which made up **67.74%** of total revenue. Female customers generated **\$75,191**, making up **32.26%** of total revenue. This suggests that male customers were responsible for a much larger share of sales in this dataset.

Business Recommendation:

The company should continue marketing to male customers since they currently generate the most revenue. However, the company should also look for ways to increase purchases from female customers, such as targeted promotions, product recommendations, or category-specific campaigns.

	gender 	revenue 	percent 
	text	numeric	numeric
1	Female	75191	32.26
2	Male	157890	67.74

Finding 2: Many discount users still spent above the average purchase amount

My SQL query found **839 customers** who used a discount but still spent more than the average purchase amount. This is important because it shows that discounts did not only attract

low-spending customers. Many customers who received discounts still made higher-value purchases.

Business Recommendation:

The company should continue using discounts, but it should focus them on customers who are likely to spend more. Discounts can be used as a strategy to encourage larger purchases instead of just lowering prices for all customers.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839		Query complete 00:00:00.096

Finding 3: Gloves, sandals, and boots had the highest average review ratings

These products had the strongest review performance compared to other items.

Business Recommendation:

The company should highlight these highly rated products in marketing campaigns, featured product sections, or recommendation lists. Since these items already have stronger customer ratings, promoting them could help increase customer trust and sales.

	item_purchased text	average_review_rating numeric
1	Gloves	3.90
2	Sandals	3.89
3	Boots	3.85
4	Hat	3.81
5	Skirt	3.81

Finding 4: Express shipping customers spent slightly more than standard shipping customers

Customers using Express shipping had an average purchase amount of **\$60.48**, while customers using Standard shipping had an average purchase amount of **\$58.46**. The difference is not huge, but it suggests that customers choosing faster shipping may be slightly more willing to spend.

Business Recommendation:

The company could promote Express shipping as a premium option, especially during busy seasons or for higher-value purchases. It may also be useful to test free or discounted Express shipping for customers who reach a certain spending amount.

	shipping_type text	average_purchase_amount numeric
1	Express	60.48
2	Standard	58.46

Finding 5: Non-subscribers generated more total revenue, but subscribers spent almost the same on average

Non-subscribers generated more total revenue because there were many more of them. However, the average spending between subscribers and non-subscribers was almost the same.

Business Recommendation:

The company should work on increasing subscription sign-ups because subscribers do not spend much less than non-subscribers. If more customers become subscribers, total revenue could increase while also building a more loyal customer base.

	subscription_status text	total_customers bigint	total_spent numeric	average_spent numeric
1	No	2847	170436	59.87
2	Yes	1053	62645	59.49

Finding 6: Some products depend heavily on discounts

Certain products like hats, sneakers, and coats are more often purchased with discounts than others.

Business Recommendation:

The company should review whether these discounts are actually improving sales or just reducing profit. Products like hats, sneakers, coats, sweaters, and pants may need better pricing strategies, seasonal promotions, or bundle offers instead of frequent discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Finding 7: Most customers were loyal customers

Most customers were classified as loyal, meaning they had a higher number of previous purchases.

Business Recommendation:

Since loyal customers make up the largest group, the company should create or improve loyalty programs. Rewards, exclusive offers, early access to products, or personalized recommendations could help keep these customers engaged.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	Returning	701
3	New	83

Finding 8: Clothing and accessories had several of the most purchased products

The top products by category showed that:

- Accessories: Jewelry, sunglasses, and belts were the top items.
- Clothing: Blouses, pants, and shirts were the top items.
- Footwear: Sandals, Shoes, and Sneakers were the top items.
- Outerwear: Jacket and Coat were the top items shown.

Business Recommendation:

The company should make sure these high-demand products are well-stocked and easy to find on the website or in-store. These items could also be used in product bundles or seasonal campaigns.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Finding 9: Most repeat buyers were not subscribed

Among customers with more than 5 previous purchases, **2,518 repeat buyers were not subscribed**, while **958 repeat buyers were subscribed**.

This means many loyal or repeat customers have not joined the subscription program.

Business Recommendation:

The company should target repeat buyers who are not subscribed. These customers already show loyalty through repeat purchases, so they may be good candidates for subscription offers, loyalty rewards, or member-only benefits.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Finding 10: Young adults contributed the highest revenue by age group

Young adults contributed the largest share of revenue, but the revenue distribution across age groups was fairly balanced.

Business Recommendation:

The company should continue targeting young adults because they produced the highest revenue. However, since all age groups contributed a similar share, marketing should not focus only on one age group. A balanced strategy with age-specific campaigns would likely work best.

	age_group text	total_revenue numeric	percent numeric
1	Young Adult	62143	26.66
2	Middle-aged	59197	25.40
3	Adult	55978	24.02
4	Senior	55763	23.92